

Chapter 2

Acids, Bases and Salts

1. Which of the following acids is present in lemon?

- A) Acetic acid
- B) Citric acid
- C) Lactic acid
- D) Formic acid

Answer: B) Citric acid

2. The acid present in vinegar is:

- A) Formic acid
- B) Hydrochloric acid
- C) Acetic acid
- D) Citric acid

Answer: C) Acetic acid

3. A base that is soluble in water is called a:

- A) Acid
- B) Alkali
- C) Salt
- D) Neutral substance

Answer: B) Alkali

4. Which of the following is a strong acid?

- A) H_2CO_3
- B) HCl
- C) CH_3COOH
- D) H_2S

Answer: B) HCl

5. Which of the following will turn red litmus paper blue?

- A) Lemon juice
- B) Vinegar
- C) Baking soda solution
- D) Orange juice

Answer: C) Baking soda solution

6. Which of the following is not a base?

- A) NaOH
- B) KOH
- C) $\text{Ca}(\text{OH})_2$
- D) NH_4Cl

Answer: D) NH_4Cl

7. The chemical name of baking soda is:

- A) Sodium carbonate
- B) Sodium hydrogen carbonate
- C) Calcium carbonate
- D) Ammonium carbonate

Answer: B) Sodium hydrogen carbonate

8. The chemical formula of washing soda is:

- A) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- B) NaHCO_3
- C) NaOH
- D) $\text{Na}_2\text{CO}_3 \cdot 2\text{H}_2\text{O}$

Answer: A) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

9. Which one of the following will show acidic behaviour?

- A) NaCl
- B) NH_4Cl
- C) Na_2CO_3
- D) KOH

Answer: B) NH_4Cl

10. The pH of a neutral solution at 25°C is:

- A) 1
- B) 5
- C) 7
- D) 14

Answer: C) 7

11. The pH of lemon juice is around:

- A) 2
- B) 7
- C) 9

D) 10

Answer: A) 2

12. Which of the following gives relief from acid indigestion?

A) Sodium hydroxide

B) Baking soda

C) Common salt

D) Vinegar

Answer: B) Baking soda

13. Which of the following is a natural indicator?

A) Phenolphthalein

B) Methyl orange

C) Litmus

D) All of these

Answer: C) Litmus

14. The acid present in the stomach is:

A) H_2SO_4

B) HCl

C) HNO_3

D) H_2CO_3

Answer: B) HCl

15. When an acid reacts with a base, the product formed is:

A) Salt only

B) Water only

C) Salt and water

D) Acid and salt

Answer: C) Salt and water

16. Which of the following can be used to check the pH of a solution?

A) Blue litmus paper

B) Red litmus paper

C) Universal indicator

D) All of these

Answer: C) Universal indicator

17. The chemical formula of bleaching powder is:

- A) CaOCl_2
- B) CaCl_2
- C) Ca(OH)_2
- D) CaCO_3

Answer: A) CaOCl_2

18. Which of the following salts is used to remove permanent hardness of water?

- A) Baking soda
- B) Washing soda
- C) Bleaching powder
- D) Common salt

Answer: B) Washing soda

19. What is the nature of distilled water?

- A) Acidic
- B) Basic
- C) Neutral
- D) Depends on temperature

Answer: C) Neutral

20. Which of the following is used in the manufacture of soaps?

- A) NaOH
- B) NH_4OH
- C) Ca(OH)_2
- D) K_2CO_3

Answer: A) NaOH

21. Which acid is present in tamarind?

- A) Tartaric acid
- B) Acetic acid
- C) Formic acid
- D) Citric acid

Answer: A) Tartaric acid

22. The pH value less than 7 indicates:

- A) Neutral solution
- B) Basic solution
- C) Acidic solution

D) None of these

Answer: C) Acidic solution

23. Which gas is produced when acid reacts with metal?

A) Oxygen

B) Hydrogen

C) Carbon dioxide

D) Nitrogen

Answer: B) Hydrogen

24. The common salt used in food is chemically known as:

A) Sodium carbonate

B) Sodium chloride

C) Sodium bicarbonate

D) Potassium chloride

Answer: B) Sodium chloride

25. Which of the following salts is used for removing moisture from air?

A) NaCl

B) CaCl_2

C) Na_2CO_3

D) NaHCO_3

Answer: B) CaCl_2

26. Tooth decay starts when pH of mouth falls below:

A) 7

B) 6

C) 5.5

D) 4.5

Answer: C) 5.5

27. Which of the following statements is true about acids?

A) They turn blue litmus red

B) They turn red litmus blue

C) They taste bitter

D) They are soapy to touch

Answer: A) They turn blue litmus red

28. When CO_2 is passed through lime water, it turns milky due to the formation of:

- A) CaCO_3
- B) Ca(OH)_2
- C) CaO
- D) $\text{Ca(HCO}_3)_2$

Answer: A) CaCO_3

29. Which of the following salts is used in cold packs?

- A) Ammonium nitrate
- B) Sodium chloride
- C) Calcium sulphate
- D) Sodium carbonate

Answer: A) Ammonium nitrate

30. Plaster of Paris is obtained by heating gypsum at:

- A) 100°C
- B) 120°C
- C) 150°C
- D) 200°C

Answer: C) 150°C
